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Roll No

EC/EI/IC-404 (GS)
B.E. IV Semester Examination, June 2020
Grading System (GS)
Linear Integrated Circuits and Its Applications
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Write short notes on the following :
 - i) Slew rate
 - ii) Op-Amp open loop configurations
 - iii) Voltage transfer curve of Op-Amp
2. a) What happens when the common terminal of V^+ and V^- sources is not grounded?
b) Define Virtual Ground concept.
3. Explain Chebyshev filters. Write any three differences between Chebyshev and band reject filters.
4. Draw the basic inverting amplifier with an input resistance R_1 and feedback resistance R_f . Assuming the OP-AMP to be ideal, calculate the voltage gain of the inverting amplifier.
5. Calculate the voltage gain the input resistance, the output resistance and the bandwidth for a differential amplifier configuration.
6. Write short notes on the following :
 - i) PLL
 - ii) Fixed and adjustable voltage regulator
 - iii) Frequency shift keying

OR

For a Hartley oscillator, if a self inductance of the two coils are $L_1 = 100\text{mH}$, $L_2 = 1\text{mH}$ and mutual inductance between the two coils is $20\text{ }\mu\text{H}$. Find the frequency of oscillations at its output for a capacitor of 20 pF .

7. a) Explain full wave precision rectifier with neat diagram and waveforms.
b) Explain voltage to current converter and current to voltage converter with neat circuit diagram.

OR

Write short notes on:

- a) TPS40200
 - b) AGC
8. Write short notes on (any two):
 - a) AVC using Op-Amp
 - b) Schmitt Trigger
 - c) Self-Tuned Filters
 - d) Summing Amplifier
 - e) Isolation Amplifier
